

Windows CMD Essentials — Instructor Guide

Every challenge, answer, and accepted variant · generated 2026-08-18 from the live pack data

Quick reference

Set your own site URL and class password before handing this out. Admin handle: `warden` .

Handles: 3–20 characters, letters/numbers/underscore/hyphen, **claimed once only**.

27 challenges · 535 points total · flags look like `FLAG{ABCD2345EFGH}` and are **different for every student**.

How the difficulty fades — Act I briefs show the whole command, and the free hint repeats it. From Act II the brief names the tool but not its flags or arguments, and the exact line moves into a hint that costs XP. By the last act the brief states only the objective. A student who never buys a hint has genuinely recalled every command. Hint spend is visible in the instructor dashboard, so it reads as a difficulty signal rather than as cheating.

If a student says "it's not working"

Symptom	Cause & what to say
"My flag is rejected"	Flags are per-student — a classmate's flag never validates. Tell them to click the green flag in the terminal to copy it exactly, rather than retyping.
"Handle already claimed"	Someone took it, or they registered before and cleared their browser. Pick a new handle. (To free one, delete that player in Netlify Blobs.)
"I ran the command, nothing happened"	Check the blue path in their prompt — they may be in a different directory. Most challenges accept short or full paths, but the file must exist from where they stand. <code>cd ~</code> returns home.
"It says command not found"	If it's a real command we don't simulate (<code>top</code> , <code>nano</code>), the game says so explicitly. Genuine typos say "command not found".
"I don't know which command to use"	Intended from Act II on. Point them at the free hint first, then the Reference tab. The costed hint is the safety net, not a failure.
"The act is locked"	Each act opens after solving all but one of the previous act's challenges.
"Nothing printed"	Correct for <code>cd</code> and redirects. The game prints "(no output — in a shell, silence usually means success)".
Lost all progress	Progress is tied to the browser token. Same browser = resumes automatically. Different machine = new handle needed.

Act structure & unlocking

Act	Teaches	Challenges	Points	Unlock
Act I: Navigation & Inspection	CMD directory traversal, DIR switches, TYPE, and TREE	10	145	Always open
Act II: File Management & Batch	COPY, MOVE, REN, DEL, ATTRIB, and %ENVIRONMENT% variables	10	200	Solve 8 of 10 in the prior act
Act III: Filtering & Diagnostics	FINDSTR, FIND, pipelines, TASKLIST, IPCONFIG, and CERTUTIL	7	190	Solve 8 of 10 in the prior act

Students may skip **one** challenge per act and still advance.

Act I: Navigation & Inspection

CMD directory traversal, DIR switches, TYPE, and TREE

Where Am I? (CD)

10 pts · w1-cd-pwd · OK

Task: In Windows CMD, running `cd` with no arguments prints the current directory. Run `cd` to see where you stand.

Answer: `cd`

Completion: Auto-completes on: commandMatches

Hint 1 (free): Type `cd` and press Enter.

Also accepted: `chdir`

Directory Listing (DIR)

10 pts · w1-dir · OK

Task: Use `dir` to list files and subdirectories in the current folder.

Answer: `dir`

Completion: Auto-completes on: commandMatches

Hint 1 (free): Type `dir`.

Also accepted: `dir .`

Show All Files (DIR /A)

15 pts · w1-dir-a · OK

Task: Windows hides a file with an attribute letter (H for Hidden) rather than a leading dot. Make `dir` show every file, hidden ones included.

Answer: `dir /a`

Completion: Auto-completes on: commandMatches

Hint 1 (free): `dir` takes switches that start with a forward slash. The one for attributes is `/a`.

Hint 2 (-5 XP): Run `dir /a`.

Also accepted: `dir /a:h`, `dir /ah`

Bare Listing (DIR /B)

15 pts · w1-dir-b · OK

Task: Produce a clean listing of this folder: names only, with no header, no sizes and no summary line. Use `dir`.

Answer: `dir /b`

Completion: Auto-completes on: commandMatches

Hint 1 (free): The `dir` switch for a bare listing is `/b`.

Hint 2 (-5 XP): Run `dir /b`.

Recursive Search (DIR /S)

20 pts · w1-dir-s · OK

Task: Search the `Documents` folder AND every folder inside it in one go. Use `dir`.

Answer: `dir /s Documents`

Completion: Auto-completes on: `commandMatches`

Hint 1 (free): The `dir` switch that recurses into subfolders is `/s`.

Hint 2 (-5 XP): Run `dir /s Documents`.

Also accepted: `dir /s`, `dir /s .`

Enter Documents (CD)

15 pts · w1-cd-nav · OK

Task: Use `cd Documents` to move into the Documents directory.

Answer: `cd Documents`

Completion: Auto-completes on: `cwdls`

Hint 1 (free): Type `cd Documents`.

Also accepted: `cd .\Documents`, `cd C:\Users\Student\Documents`

Climb to Parent (CD ..)

15 pts · w1-cd-parent · OK

Task: Use `cd ..` from `Documents` to move back up to `C:\Users\Student`.

Answer: `cd ..`

Completion: Auto-completes on: `cwdls`

Hint 1 (free): Type `cd ..`.

Also accepted: `cd C:\Users\Student`

Display File Content (TYPE)

15 pts · w1-type · OK

Task: Print the contents of `Documents\notes.txt` to the prompt. Windows has no `cat`.

Answer: `type Documents\notes.txt`

Completion: Auto-completes on: `commandMatches`

Hint 1 (free): The cmd.exe command that prints a file is `type`.

Hint 2 (-5 XP): Run `type Documents\notes.txt`.

Also accepted: `type Documents/notes.txt`

Directory Tree (TREE)

15 pts · w1-tree · OK

Task: Display this directory and everything under it as a visual hierarchy rather than a flat list.

Answer: `tree`

Completion: Auto-completes on: `commandMatches`

Hint 1 (free): The command is `tree`. Adding `/f` also lists the files.

Hint 2 (-5 XP): Run `tree`.

Also accepted: `tree /f`

Locate Binaries (WHERE)

15 pts · w1-where · OK

Task: Find out where on disk the Windows command interpreter itself lives — the file `cmd` .

Answer: `where cmd`

Completion: Auto-completes on: `commandMatches`

Hint 1 (free): `where` searches the PATH for a named program, the way `which` does on Linux.

Hint 2 (–5 XP): Run `where cmd` .

Also accepted: `where cmd.exe`

Act II: File Management & Batch

COPY, MOVE, REN, DEL, ATTRIB, and %ENVIRONMENT% variables

Copy Files (COPY)

20 pts · w2-copy · OK

Task: Make a duplicate of `Documents\notes.txt` called `Documents\notes_backup.txt`.

Answer: `copy Documents\notes.txt Documents\notes_backup.txt`

Completion: Auto-completes on: `fileExists`

Hint 1 (free): The cmd.exe command is `copy`. It takes the source path first, then the destination.

Hint 2 (–10 XP): Run `copy Documents\notes.txt Documents\notes_backup.txt`.

Move Files (MOVE)

20 pts · w2-move · OK

Task: Move `Desktop\todo.txt` into the `Documents` folder, keeping the same filename.

Answer: `move Desktop\todo.txt Documents\todo.txt`

Completion: Auto-completes on: `fileExists`

Hint 1 (free): The cmd.exe command is `move`. Source first, destination second.

Hint 2 (–10 XP): Run `move Desktop\todo.txt Documents\todo.txt`.

Rename Files (REN)

20 pts · w2-ren · OK

Task: Rename `Desktop\Shortcut.lnk` to `AppShortcut.lnk`. It must stay on the Desktop.

Answer: `ren Desktop\Shortcut.lnk AppShortcut.lnk`

Completion: Auto-completes on: `fileExists`

Hint 1 (free): The cmd.exe command is `ren`. Give it the existing path, then the NEW NAME only — not a second full path.

Hint 2 (–10 XP): Run `ren Desktop\Shortcut.lnk AppShortcut.lnk`.

Also accepted: `rename Desktop\Shortcut.lnk AppShortcut.lnk`

Delete Files (DEL)

20 pts · w2-del · OK

Task: Delete the file `Downloads\archive.zip`.

Answer: `del Downloads\archive.zip`

Completion: Auto-completes on: `commandMatches`

Hint 1 (free): The cmd.exe command is `del`. There is no Recycle Bin from the command prompt.

Hint 2 (–10 XP): Run `del Downloads\archive.zip`.

Also accepted: `erase Downloads\archive.zip`

Create Directories (MD / MKDIR)

20 pts · w2-md · OK

Task: Create a new folder called `Projects` inside `C:\Users\Student`.

Answer: `md Projects`

Completion: Auto-completes on: `dirExists`

Hint 1 (free): `cmd.exe` accepts either `md` or the longer `mkdir`.

Hint 2 (–10 XP): Run `md Projects`.

Also accepted: `mkdir Projects`

Remove Directories (RD / RMDIR)

20 pts · w2-rd · OK

Task: Remove the `Projects` folder you just created.

Answer: `rd Projects`

Completion: Auto-completes on: `commandMatches`

Hint 1 (free): `cmd.exe` accepts either `rd` or the longer `rmdir`. The folder must be empty.

Hint 2 (–10 XP): Run `rd Projects`.

Also accepted: `rmdir Projects`

Inspect File Attributes (ATTRIB)

25 pts · w2-attrib · OK

Task: `Documents\hidden_config.ini` does not appear in a plain `dir`. Show which attributes are set on it.

Answer: `attrib Documents\hidden_config.ini`

Completion: Auto-completes on: `commandMatches`

Hint 1 (free): `attrib` reports a file's attribute letters. H is Hidden, R is Read-only, S is System.

Hint 2 (–10 XP): Run `attrib Documents\hidden_config.ini`.

Also accepted: `attrib Documents/hidden_config.ini`

Environment Variables (%VAR%)

20 pts · w2-env-var · OK

Task: `cmd.exe` wraps an environment variable in percent signs, like `%NAME%`. Print the value of your `USERPROFILE` variable.

Answer: `echo %USERPROFILE%`

Completion: Auto-completes on: `commandMatches`

Hint 1 (free): `echo` prints text, and the shell substitutes the variable before `echo` sees it.

Hint 2 (–10 XP): Run `echo %USERPROFILE%`.

Setting Variables (SET)

25 pts · w2-set · OK

Task: Create a new environment variable called `MY_VAR` and give it the value `123`.

Answer: `set MY_VAR=123`

Completion: Auto-completes on: commandMatches

Hint 1 (free): The cmd.exe command is `set`. Write the assignment with no spaces around the equals sign.

Hint 2 (-10 XP): Run `set MY_VAR=123`.

Clear the Console (CLS)

10 pts · w2-cls · OK

Task: The screen is cluttered. Clear it.

Answer: `cls`

Completion: Auto-completes on: commandMatches

Hint 1 (free): The cmd.exe command is `cls`, short for clear screen.

Hint 2 (-10 XP): Run `cls`.

Act III: Filtering & Diagnostics

FINDSTR, FIND, pipelines, TASKLIST, IPCONFIG, and CERTUTIL

Pattern Search (FINDSTR)

25 pts · w3-findstr · OK

Task: Show every line in `logs\eventlog.txt` that contains the phrase `EVENT ID`. Windows has no `grep`.

Answer: `findstr "EVENT ID" logs\eventlog.txt`

Completion: Auto-completes on: commandMatches

Hint 1 (free): The cmd.exe search command is `findstr`. Quote the phrase, because it contains a space.

Hint 2 (–10 XP): Run `findstr "EVENT ID" logs\eventlog.txt`.

Also accepted: `findstr 'EVENT ID' logs\eventlog.txt`

Case-Insensitive FINDSTR (/I)

25 pts · w3-findstr-i · OK

Task: Find every mention of `defender` in `logs\eventlog.txt`, no matter how it is capitalised.

Answer: `findstr /i "defender" logs\eventlog.txt`

Completion: Auto-completes on: commandMatches

Hint 1 (free): `findstr` switches start with a forward slash. The one that ignores case is `/i`.

Hint 2 (–10 XP): Run `findstr /i "defender" logs\eventlog.txt`.

Also accepted: `findstr /i "defender" logs\eventlog.txt`

Count Lines (FIND /C /V "")

30 pts · w3-find-count · OK

Task: Count how many lines `logs\firewall.log` contains. cmd.exe has no `wc`, so the standard trick is to print the file and pipe it into a counter that counts every line NOT matching an empty pattern.

Answer: `type logs\firewall.log | find /c /v ""`

Completion: Auto-completes on: commandMatches

Hint 1 (free): `type` prints the file, `|` pipes it onward, and `find` counts with `/c`. Add `/v ""` so it counts the lines that do not match nothing — that is, all of them.

Hint 2 (–10 XP): Run `type logs\firewall.log | find /c /v ""`.

Instructor note: The `find /c /v ""` idiom is genuinely unguessable — expect most of the class to buy the hint.

Process List (TASKLIST)

25 pts · w3-tasklist · OK

Task: Show every process running on this machine, with its memory use.

Answer: `tasklist`

Completion: Auto-completes on: commandMatches

Hint 1 (free): The cmd.exe equivalent of Task Manager is a single command: `tasklist`.

Hint 2 (–10 XP): Run `tasklist`.

Network Configuration (IPCONFIG)

25 pts · w3-ipconfig · OK

Task: Show this machine's IP address and network adapter configuration.

Answer: `ipconfig`

Completion: Auto-completes on: `commandMatches`

Hint 1 (free): The `cmd.exe` command is `ipconfig`. Adding `/all` shows much more detail.

Hint 2 (-10 XP): Run `ipconfig`.

Also accepted: `ipconfig /all`

System Information (SYSTEMINFO)

25 pts · w3-systeminfo · OK

Task: Display the operating system build, installed memory and hardware summary for this machine.

Answer: `systeminfo`

Completion: Auto-completes on: `commandMatches`

Hint 1 (free): One command reports the whole system summary: `systeminfo`.

Hint 2 (-10 XP): Run `systeminfo`.

Cryptographic Hash (CERTUTIL)

35 pts · w3-certutil · OK

Task: Compute the MD5 checksum of `Documents\data.csv` for a chain-of-custody record. Windows has no `md5sum`; the built-in tool is `certutil`.

Answer: `certutil -hashfile Documents\data.csv MD5`

Completion: Auto-completes on: `commandMatches`

Hint 1 (free): `certutil` needs the `-hashfile` switch, then the file path, then the algorithm name in capitals.

Hint 2 (-10 XP): Run `certutil -hashfile Documents\data.csv MD5`.

Also accepted: `certutil -hashfile Documents/data.csv MD5`

Accepted command variants

Auto-completing challenges accept the natural variations a beginner types, so a correct command is never silently ignored:

- Optional quotes around a path
- Path prefixes: `./Documents`, `~/Documents`, or the full absolute path
- Short paths when already inside the folder
- Flag spacing: `head -n 5`, `head -n5`, `head -5`
- Windows: forward or back slashes, any capitalization (`md5` or `MD5`)
- Trailing spaces are ignored everywhere

Teaching notes

- **The Coach** (toggle in the terminal title bar) explains every command and error in plain language. Explanations fade after the second use of a command. Leave it ON for beginners.
- **Reference tab** holds a manual page for every command, and tells students the real-shell equivalent. Point stuck students there before they buy a hint.
- **Unsimulated tools** reply with what they are, rather than a bare "command not found".

- **Hints:** the first is free and conceptual; the one that gives the exact command costs XP. Hint use is visible in the instructor dashboard as a signal of where the class is struggling.
- **Watch the stuck points.** The dashboard flags any challenge with a solve rate under 35%. Now that briefs no longer contain the answer, that number means something: it marks a concept the class did not get, not a typing error.

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